

A NOTE ON THE CHARACTERIZATION OF THE EXPONENTIAL DISTRIBUTION BY THE COVARIANCE OF SPACINGS

J.W.H. Swanepoel: Department of Statistics and Operations Research, Potchefstroom University for CHE, Potchefstroom, 2520, Republic of South Africa

Key words: Characterization, covariance, exponential distribution, spacings.

Summary: Suppose $X_1, X_2, \dots, X_n$ is a random sample from a continuous distribution function F and let $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ be the corresponding order statistics. A characterization of the exponential distribution based on the covariance of $X_{(1)}$ and $\sum_{i=2}^n (X_{(i)} - X_{(1)})/(n-1)$ is presented and links with other clearly related results are established.

1. INTRODUCTION

Let $X_1, X_2, \dots, X_n$ be $n \geq 2$ mutually independent random variables with a common continuous distribution function F . Let $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ denote the corresponding order statistics, and define $Z = \sum_{i=2}^n (X_{(i)} - X_{(1)})/(n-1)$. Throughout the discussion below the two-parameter exponential distribution will be denoted by F_0 , i.e.

$$F_0(x) = 1 - \exp\{-(x - \theta)/\sigma\}, \quad x > \theta > -\infty, \sigma > 0 \\ = 0, \text{ elsewhere.}$$

Many results are available concerning the characterization of F_0 by means of order statistics, and for surveys of the literature in this area, we

refer the reader to Kotz (1974) and Galambos (1975). Probably one of the best known results is that by Tanis (1964) who showed that an absolutely continuous distribution function is negative exponential if and only if $X_{(1)}$ and Z are independent. A very general extension of this result, where only continuity of F is required, was presented by Rossberg (1972).

Another result is due to Dallas (1973) who showed that, among all continuous distributions having finite first moment, the exponential distribution F_0 is the unique distribution satisfying $E(Z|X_{(1)}) = E(Z)$, almost surely. Generalizations of this result were also studied by Beg and Kirmani (1974), Wang and Srivastava (1980). The latter two authors pointed out that the characterization of the exponential distribution based on the constant regression of Z on $X_{(1)}$ is not stronger than the result obtained by using the independence of Z and $X_{(1)}$, because in the latter case it is not necessary to assume the existence of the first moment of F (see Rossberg, 1972). For related results we refer to Galambos and Kotz (1978) and also Azlarov and Volodin (1986).

The purpose of this note is to define a class $\mathcal{H}$ of continuous distributions and prove that the exponential distribution F_0 is the only member of this class satisfying $\text{cov}(X_{(1)}, Z) = 0$. These results are presented in section 2. In section 3 it is shown that $\mathcal{H}$ is a large class, consisting of most known and important distributions.

2. MAIN RESULT

Let $\mathcal{H}$ denote the class consisting of all continuous distribution functions F such that

$$(2.1) \quad \int_{-\infty}^{+\infty} x^2 dF(x) < \infty,$$

$$(2.2) \quad r(x) = \frac{\int_x^{\infty} (1 - F(y)) dy}{1 - F(x)}$$

is a monotone function of x .

THEOREM. The following two conditions are equivalent:

- (i) $F = F_0$
- (ii) $\text{cov}(X_{(1)}, Z) = 0$ and $F \in \mathcal{K}$

PROOF. Wang and Srivastava (1980) proved that (see their theorem 1, page 219) $r(x) = E(Z | X_{(1)} = x)$. Now, let $s(x) = E[Z - E(Z) | X_{(1)} = x]$ and first assume that $s(x)$ is a nonconstant nondecreasing function. Then since $E(s(X_{(1)})) = 0$, there exists a constant c such that $s(x) \leq 0$ if $x \leq c$, $s(x) \geq 0$ if $x \geq c$ and $s(x) \neq 0$ for some $x \neq c$. Applying a similar argument as in lemma 2.1 of Bickel (1967) we obtain

$$\begin{aligned}
 (2.3) \quad \text{cov}(X_{(1)}, Z) &= E\{X_{(1)} E[Z - E(Z) | X_{(1)}]\} \\
 &= E[(X_{(1)} - c)s(X_{(1)})] \\
 &> 0.
 \end{aligned}$$

Similarly, if $s(x)$ is a nonconstant nonincreasing function then $\text{cov}(X_{(1)}, Z) < 0$. Also, if $s(x)$ is almost everywhere constant then trivially this constant should be zero, and consequently the first line of (2.3) implies that $\text{cov}(X_{(1)}, Z) = 0$.

The proof of the theorem now follows from the above inequalities and a characterization theorem of Dallas (1973), which states that the exponential distribution F_0 is the only distribution for which $s(x)$ (or $r(x)$) is almost everywhere constant.

3. DISTRIBUTIONS SATISFYING (2.2)

The class $\mathcal{K}$ is large and includes, among others, the following distributions: three-parameter gamma, two-parameter normal, two-parameter double exponential, two-parameter logistic, two-parameter extreme value, Pearson (type 1), Pareto (of the second kind), three-parameter Weibull, IFR and DFR life distributions. We refer the reader to Johnson and Kotz (1972), Hollander and Proschan (1984) for definitions and details of these distributions.

The above assertion will now be proved for the gamma and normal distributions, since they require different argumentation. Proofs for the other distributions are available from the author on request.

(A) The gamma distribution

Suppose the probability density of F is of the form

$$f(x) = \frac{(x - \theta)^{\alpha-1} \exp[-(x - \theta)/\sigma]}{\sigma^\alpha \Gamma(\alpha)}, \quad x > \theta > -\infty, \alpha > 0, \sigma > 0$$

$$= 0, \text{ elsewhere.}$$

To prove that $r(x)$, given in (2.2), is monotonic it is easy to see that one may assume, without loss of generality, that $\theta = 0$ and $\sigma = 1$. Define

$$H_\alpha(x) = \int_x^\infty y^{\alpha-1} e^{-y} dy$$

then, for each $x > 0$ and $\alpha > 0$, integration by parts yields

$$x^\alpha e^{-x} = H_{\alpha+1}(x) - \alpha H_\alpha(x).$$

Using this identity, it follows fairly easily that

$$(3.1) \quad r'(x) = \frac{(\alpha - 1)[H_\alpha^2(x) - H_{\alpha-1}(x)H_{\alpha+1}(x)]}{H_\alpha^2(x)}.$$

Also, applying the Schwarz inequality, we find that for each $x > 0$

$$(3.2) \quad H_\alpha^2(x) = \left\{ \int_x^\infty [y^{(\alpha-2)/2} e^{-y/2}] [y^{\alpha/2} e^{-y/2}] dy \right\}^2$$

$$< H_{\alpha-1}(x) H_{\alpha+1}(x).$$

Hence, from (3.1) and (3.2) we conclude that $r'(x) < 0$ (> 0) for all $x > 0$ if $\alpha > 1$ (< 1), and $r'(x) = 0$ if $\alpha = 1$.

(B) The normal distribution

Consider the normal distribution $F(x) = \Phi((x - \mu)/\sigma)$, $-\infty < x < \infty$, $-\infty < \mu < \infty$, $\sigma > 0$. As usual, Φ and ϕ denote the standard normal distribution function and density function respectively. We claim that $r'(x) < 0$ for all x . Once again we assume, without loss of generality, that $\mu = 0$ and $\sigma = 1$. Note that

$$r'(x) = \frac{-(1 - \Phi(x))^2 + \phi(x) \int_x^{\infty} (1 - \Phi(y)) dy}{(1 - \Phi(x))^2}$$

and denoting the numerator by $g(x)$, we find that

$$g(x) = -(1 - \Phi(x))^2 + \phi(x)\{-x(1 - \Phi(x)) + \phi(x)\}$$

which implies that $g(x) \rightarrow -1$ if $x \rightarrow -\infty$, and $g(x) \rightarrow 0$ if $x \rightarrow \infty$. Our claim above will now follow if one can prove that $g'(x) \geq 0$ for all x . Hence,

$$g'(x) = \phi(x)(1 - \Phi(x)) - x\phi(x) \int_x^{\infty} (1 - \Phi(y)) dy.$$

Clearly, $g'(x) \geq 0$ for all $x \leq 0$. For $x > 0$ we obtain

$$g'(x) \geq \phi(x)(1 - \Phi(x)) - \phi(x) \int_x^{\infty} y(1 - \Phi(y)) dy \geq 0$$

which follows from the well-known Mills inequality $y(1 - \Phi(y)) < \phi(y)$ for all $y > 0$.

ACKNOWLEDGEMENTS

The author is grateful to C.F. de Beer for his helpful discussions and constructive criticisms. This research was supported by grants from the FRD of the CSIR of South Africa.

REFERENCES

- Azlarov, T.A. & Volodin, N.A. (1986). *Characterization problems associated with the exponential distribution*, Springer-Verlag.
- Beg, M.I. & Kirmani, S.N.U.A. (1974). On a characterization of exponential and related distributions, *Austral. J. Statist.*, 16, 163-166. (Correction in *Austral. J. Statist.* (1976) 18, 85.)
- Bickel, P.J. (1967). Some contributions to the theory of order statistics, *Proc. 5th Berkeley Symp.*, 1, 575-591.
- Dallas, A.C. (1973). A characterization of the exponential distribution, *Bull. Soc. Math. Greece*, 14, 172-175.
- Galambos, J. (1975). Characterizations of probability distributions by properties of order statistics I-II, In *Statistical Distributions in Scientific Works*, 3. (G.P. Patil, S. Kotz and J.K. Ord, eds.), 71-101. D. Reidel Publishing Co., Boston.
- Galambos, J. & Kotz, S. (1978). *Characterizations of probability distributions*, Springer-Verlag, New York.
- Hollander, M. & Proschan, F. (1984). Nonparametric concepts and methods in reliability, *Handbook of Statistics, Vol. 4, Nonparametric methods*, edited by P.R. Krishnaiah and P.K. Sen, 613-655.
- Johnson, N.L. & Kotz, S. (1972). *Continuous univariate distributions, I and II*, Houghton Mifflin Co., Boston.
- Kotz, S. (1974). Characterizations of statistical distributions: a supplement to recent surveys, *Internat. Statist. Rev.*, 42, 39-66.
- Rossberg, H.-J. (1972). Characterization of distribution functions by the independence of certain functions of order statistics, *Sankhya, Ser., A* 34, 111-120.
- Tanis, E. (1964). Linear forms in the order statistics from an exponential distribution, *Ann. Math. Statist.*, 35, 270-276.
- Wang, Y.H. & Srivastava, R.C. (1980). A characterization of the exponential and related distributions by linear regression, *Ann. Statist.*, 3, 217-220.